Quarterly Infrastructure Report

The federal geodetic service operates a nationwide network of GNSS reference stations. This quarter 148 stations were upgraded to multi-frequency receivers with real-time kinematic correction streams.

Data availability across the network reached 99.2 percent. Downtime was mainly caused by power outages at three remote alpine sites.

The transformation parameters for the national coordinate reference frame were recomputed using the latest epoch. Horizontal shifts stay below two centimetres for the whole territory.

Vector Search and Retrieval

A vector store keeps high dimensional embeddings and answers nearest neighbour queries. Cosine similarity is the usual metric for text, because it ignores magnitude and compares direction only.

Hybrid retrieval combines dense semantic embeddings with sparse keyword matching such as BM25, then fuses the two ranked lists with reciprocal rank fusion. It improves recall for rare terms and acronyms.

Chunking strategy matters: overlapping windows of roughly one thousand characters preserve context across page boundaries while keeping each chunk small enough for precise retrieval.

Cadastral Data Quality

Quality control of the digital cadastre follows the ISO 19157 framework. Completeness, logical consistency and positional accuracy are measured against the register of parcel boundaries.

Positional accuracy of the surveyed boundary points is within five centimetres at the 95 percent confidence level for urban parcels.